5-Mark Questions & Answers

Q1. Explain the Von Neumann model with a neat labeled diagram.

Introduction

The Von Neumann Architecture proposed by John von Neumann in 1945 forms the foundational model for modern general-purpose digital computers. Its defining characteristic is the Stored Program Concept, where both data and program instructions share the same memory space.

Key Components

1. Central Processing Unit (CPU):

* **Co

Block Diagram

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Q2. Compare Von Neumann and Harvard architecture (tabular format, at least 5 points).

| Feature | Von Neumann Architecture | Harvard Architecture |

| :--- | :-

Q3. Explain the architecture of the 8086 microprocessor with a block diagram (BIU and EU in detail).

Introduction

The Intel 8086 is a 16-bit microprocessor with a 20-bit address bus (1 MB memory). To maximize efficiency, its internal architecture is divided into two independent, asynchronously operating units: the Bus Interface Unit (BIU) and the Execution Unit (EU).

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1. Bus Interface Unit (BIU)

* **Functions:** Handles all external bus transfers - instruction fetching, memory read/write, and I/O access.

* **Co

2. Execution Unit (EU)

* **Functions:** Decodes and executes instructions fetched from the BIU queue.

* **Co

Q4. Explain any five performance measures of computer architecture with formulas.

1. CPU Execution Time: The total time taken by the CPU to execute a program.

* `CPU

Q5. Explain the instruction cycle of the Von Neumann model with a flow diagram, including interrupt handling.

Instruction Cycle Stages

1. Fetch: Read instruction from memory at address given by Program Counter (PC). Load into Instruction Register (IR). Increment PC.

2. Deco

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Q6. Classify and explain the different categories of the 8086 instruction set with examples.

1. Data Transfer Instructions: Move data between registers, memory, and I/O ports.

* *Exa

Q7. Explain any five addressing modes of the 8086 with syntax, effective address calculation, and examples.

1. Immediate Addressing Mode: Operand is a constant embedded in the instruction.

* *Syn

Q8. Explain data transfer instructions (MOV, XCHG, LEA, PUSH, POP) with examples.

1. MOV (Move): Copies data from source to destination operand.

* *Exa

Q9. Explain control transfer instructions (JMP, CALL, RET, LOOP, Jcc) with examples.

1. JMP (Unconditional Jump): Transfers control unconditionally to target address.

* *Exa

Q10. Explain logical instructions (AND, OR, XOR, NOT, TEST, CMP) with the flags affected.

1. AND: Bitwise logical AND. Clears CF and OF. Sets ZF, SF, PF based on result.

* *Exa

Q11. Given clock rate, CPI, and instruction count, calculate CPU execution time and MIPS (numerical problem).

Problem Statement

A processor runs at a Clock Rate of 2 GHz (2 x 10^9 Hz). A program executes 10 x 10^6 instructions (Instruction Count) with an average CPI of 2.5.

Calcula

Solution

1. Calculate CPU Execution Time:

Equation:

`CPU Ex`

Equation:

`CPU Ex`

Equation:

`CPU Ex`

2. Calculate MIPS Rating:

Equation:

`MIPS =`

Equation:

`MIPS =`

Alternatively:

Equation:

`MIPS =`
